

eyes, a hierarchical organization and extraction of information to convey to the brain and then process the information.

Several factors operate to shape and sometimes distort perception. These factors can reside in the perceiver, in the object or target being perceived, or in the context of the situation in which the perception is made. This happens for example at the level of the retina for some type of illusions.

We see things as we are, and this is what we really call perception. It's not just the process of receiving the information but also making sense of this piece of information, this is how experience works. Perception can be influenced by several things, like physiological state (food when I'm hungry) or the context (crosta o crumble di pane). It's plenty of things that can alter our perception of things (just like blue risotto).

Organization is the process of placing selected stimuli in a framework, providing categorical recognition.

The f-MRI

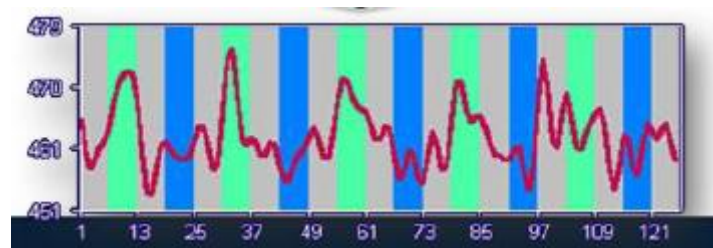
MRI studies brain anatomy, f-MRI studies the functioning. MRI is primarily using one image with high resolution, for f-MRI we acquire different images in time (every 2 sec for 5 minutes) with low resolution. Comparing it with PET imaging we are adding the temporal component.

Nuclear magnetic resonance is based the electromagnetic stimulation that deviates, flips the proton, and once the stimulus is over you let it realign with the magnetic field with a specific relaxation time, that influences different aspects.

We are dealing with endogenous contrast, the signal that we are measuring is the BOLD signal (**Blood Oxygenation Level Dependent signal**). When you have stimulation, you have a higher consumption of glucose and higher blood flow, but the flux of oxygen is not proportional to the amount of oxygen that the brain receives (luxury perfusion, it arrives more blood than needed).

Oxygen and carbon dioxide are bonded to one protein in the red blood cells, oxyhemoglobin, that has different magnetic properties, is slightly diamagnetic (attractive) while deoxyhemoglobin is paramagnetic. So, you have an indigenous tracer that has two different behaviors, and the T2 (relative to the interaction with hemoglobin) is reducing the signal when there is a higher concentration of deoxyhemoglobin (slightly attractive), so that in resting condition you have an unbalance between the arterial side and the venous side.

When I have a stimulus that activates the brain, I have a lot of oxygenated blood (more blood flow than oxygen consumption), and while in the arterial side there are no changes in the venous side something happens. This is because some oxygenated blood gets into the venous side, and becomes more diamagnetic, increasing the signal also in this side.



Cleaned signal in which we can see the stimulus.

BOLD signal is 12000-15000 ms, not related to action potential. The phenomenon I want to record is delayed by 4-6 seconds in respect to the moment the phenomenon occurs, because you must wait for the blood flow.

While in PET (8-10 volumes per subject) you are creating a difference between two responses (higher or lower) to some stimulus, in f-MRI (800-1000 volumes per subject) we are contrasting two temporal correlations to the stimulus you are providing. 2-3 ml, so higher number of voxels, higher number of information we can process.

There can be different strategies for neural recruitment. You may have very solid responses (block responses) for instance, but after few seconds you get bored, so you can play with different intervals. Example:

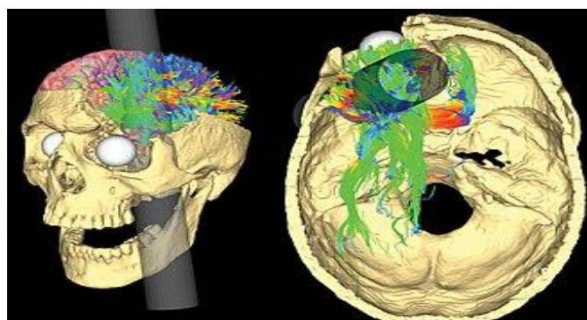
- Yellow → Faces
- Blu → Fruits



Lesson 15/10/2025 (The study of the brain and brain function)

We have many different techniques able to study the brain. When we study humans we use non-invasive ones, always considering that we are dealing with patients, the most used are the electroencephalography (the cost depends on the # of electrodes, max 256, generally cheap) and f-MRI, they are the cheapest and most accessible.

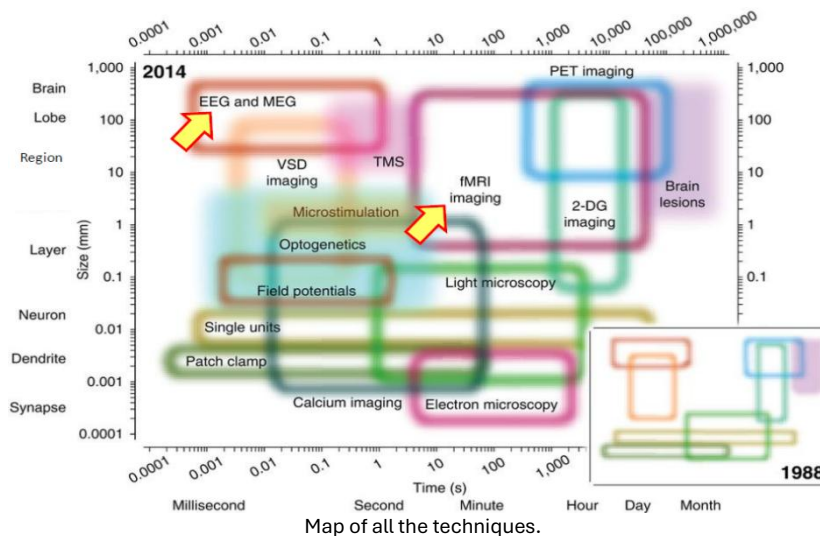
Example: The case of Phineas Gage. Lesion in the frontal cortex and a critical change in the behavior (no control and rage problem).



The studies on brain lesions are not good in general.

- Brain lesions are rarely selective, when you have a brain lesion it often involves different parts of the brain at the same time (inflammatory reactions, edema around the lesion, affect other lesions).
- Symptoms can change with time, after the damage you can recover, the edema decreases, the plastic processes that take place in the brain allow it.
- Sometimes the deficit is not identifiable: there are patients that have damage in the brain and can lose the ability to perceive colors, but depending on the type of lesion that they have they may not realize it.

This is why brain imaging became so important. We can get a more precise idea of what happened in the brain without relying on lesions and we can use some manipulations of behavior or of the experimental conditions to see what changes in the brain of the participants.



In the y axis you have the spatial resolution of the imaging technique, in the x axis the temporal one. All the techniques in the bottom part of the y axis are not accessible to humans, only animals. In humans you can't have access to the individual neuron. Most of our techniques allow us to see in the level of regions (centimeters/millimeters), or the level of the brain overall and of the lobes.

EEG and MEG have comparable temporal and spatial resolution, with a very good temporal resolution (order of ms, detect neural activity real time), but they have very bad spatial resolution. You can't see where the changes come from.

f-MRI has a much better spatial resolution, does not track neuronal and brain activity, only blood flow, a consequence of brain activity. If you want, you can apportion changes in oxygen and in blood flow that are not related to brain activity. In addition, the main component that causes neurovascular coupling is the astrocyte, not the neuron. This imaging technique has a low temporal resolution, you acquire an image of the brain in 2-3 seconds (really slow), good to get an idea of which regions respond to a certain task, but not to discover for example which region responds first.

If you want to have something that interacts with the patient you can use what is called neurofeedback. A type of protocol (software, program) which changes the stimulus in the participant based on the changes in brain activity (used to control a robotic hand, not possible with f-MRI with the temporal limitations).

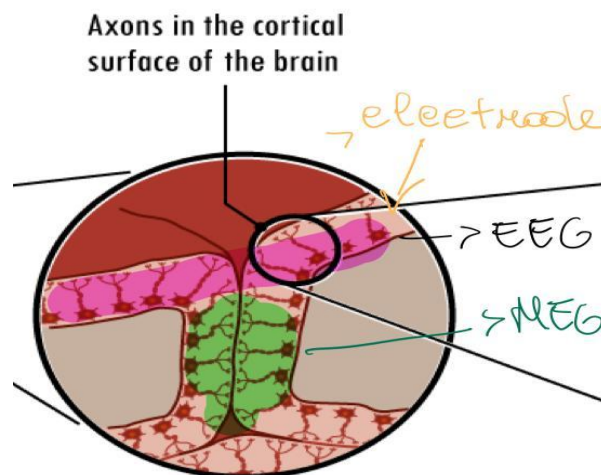
The EEG is different, the main point is that you are looking at the neural activity, you see changes related to pyramidal neurons activation, perpendicular to the surface of the cortex. The difference in potential related to asynchronous action potential is measured on the scalp. But the important thing about the EEG is that the electrical activity is volume conducted, you get a potential that is a difference between the body of the cell and the neurites. It represents a problem, because the electrical activity propagates through the tissue in all directions. Volume conduction:

- Good because regardless of where the activity is generated some electrons on the scalp will be able to get it, propagating through the tissues to the scalp.

- Bad because you get dispersion through the path and you have to amplify the signal.

The difference between MEG and EEG is that the last one is based on the detection of a signal that propagates in the same way as for the source, meaning that the signal that we detect is produced by the pyramidal neurons. For the MEG you are looking at the magnetic field produced by the current, and the magnetic field is perpendicular to the current that is produced, so if the neurons are perpendicular to the surface of the scalp, then the magnetic field will be directed in the perpendicular way, which means that the sensors that are exactly above the scalp will not detect the signal produced by that neurons.

MEG and EEG are related because they exploit the same kind of generator, but they could not necessarily produce the same signal, because they look at slightly different perspectives. EEG is a little better, because it can detect all the electrical activity produced by all the neurons.



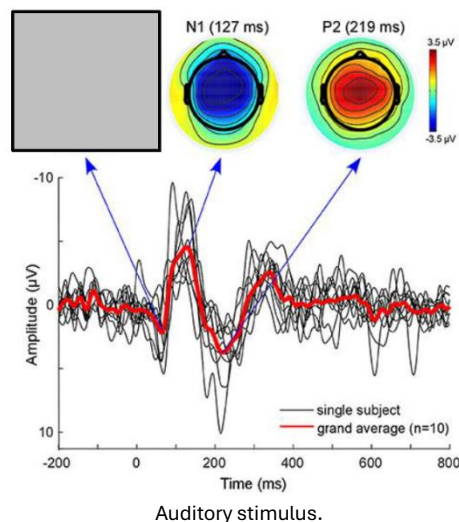
Example of signal detection, MEG can detect only the signals produced by sulcus neurons. EEG can detect all.

Some people prefer MEG, this because the spatial resolution is slightly better, because the signal is not distorted or propagated it is more precise. True based on models but not so much in reality. MEG is accessible by very few laboratories, less portable and less versatile in the use of the machine itself.

Different numbers of channels based on the use of the machine. For clinical views the laboratories use up to 21 electrodes, a number that is sufficient, but eventually for commercial EEG also only 7 sensors placed on the forehead. The problem with the EEG is that the contacts must have good contact with the scalps, you need lowest possible impedance (conductive gel, even 1h 30 just to prepare the subject), with 256 electrodes it could be difficult for a patient in an experiment. The lack of hair does not improve the conductivity but lowers it, because the scalps become thicker. The boundary between what is called standard EEG and High-Density EEG is the 64 electrodes configuration.

There could be also a net, immersed in water and soap, but also gel can be replaced with sponges. The subject in this way can be wet and once the sponge dries up the signal is lost. It's a good solution depending on what you do, there are cases where you have to be fast (children, people with disorders). Dry electrodes exist but their signal is very bad, noisy, good for piloting and testing, research is going towards implanted electrodes (many patients don't want the implant).

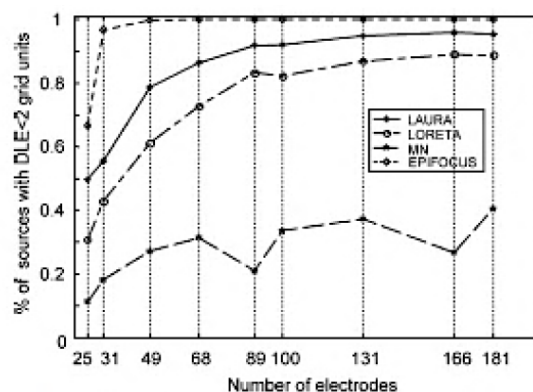
The EEG has bad spatial resolution, but very good temporal resolution.



We are not able to know where the activity started, we know what type of signal it is based on the form of it. We rely on the electrodes positioning and then a sum of the different signals.

The spatial resolution is constraint with some factor; one is the algorithm, reconstruction approaches applied can have slightly different assumptions and performances. You also need two other things, the MRI of the participant, to reconstruct the brain anatomy of the patient and have a segmentation of the tissues, to get a very accurate map of how the neural activity propagates in the brain. The second thing is that you need the exact position of the electrodes on the scalp, and then you have to align the electrodes position with the MRI. With these information's we can get a very accurate model of the brain, and consequently a great spatial resolution comparable to the one of MRI (mm).

The gain of having many electrodes, just like in the high-density EEG, is that if you want source reconstruction and you have a large coverage of the brain you have a more accurate reconstruction.



Accuracy of localization of a focus epilepticus (using source modelling)

We can see that error in source localization decreases with the addition of electrodes on the scalp. We don't use more than 256 electrodes because the time preparing the patients critically increases and the more electrodes you have there is the possibility that an electrode making contact with the gel of another one can create a bridge and the current would be the same for the two (you end up with a lower spatial resolution).

When you choose a technique to study the brain you need to think of what you want to know (spatial distribution of the processes in the brain?). EEG is cheaper and more accessible, also used in movable conditions even though there are a lot of artifacts. The main problem is that the subject is alive. Eventually you can program the use of a machine (ex. drone) relying on the eyes or muscle signal artifacts, not on the brain.

A lot of people prefer f-MRI, but the problem is what you need to get. This type of imaging technique is very fancy, if you need high temporal resolution you can't rely on it. Typically to have a more informative view you have to merge different techniques, also transcranial magnetic stimulation. There is no perfect technique, it all depends on what you want to get.

There is a position (near corona) in which there are no muscles, so no artifacts, but the speed of reaction is very low. With every heartbeat the EEG becomes very noisy, this is because the brain (the head) slightly moves inside the scanner, and the magnetic field produces a current in the EEG system.

You can get neural navigation (jump in the dark, you don't know where you are stimulating) by integrating EEG and TMS, getting the effect of the stimulation inducted, through the EEG.

Important rule in neuroscience (Garbage In Garbage Out), meaning that if the data is collected in a bad way there is no method of analyzing that can improve the quality of the signal.

Consciousness (application of the methods described)

Consciousness is something that was not considered a science at first, because of a big debate on what parts of the brain are that makes us conscious. One theory says that any complex system can be conscious (the problem is only how conscious you are), the other says that without the frontal cortex we can't have consciousness (some animals have very small frontal cortex, or even newborns).



To say that someone is conscious you have to say what you mean by it, in this case she was conscious, having a dream. Is a person in vegetative state (coma) conscious? Seeing him you wouldn't say so, but actually he is so, and you can verify it by using the f-MRI making questions and seeing the answers.